

MAK COLLEGIATE

Chapter # 15

HOMEOSTASIS

ADMIN:

ABU HURAIKA AHMAD



ZOOLOGY



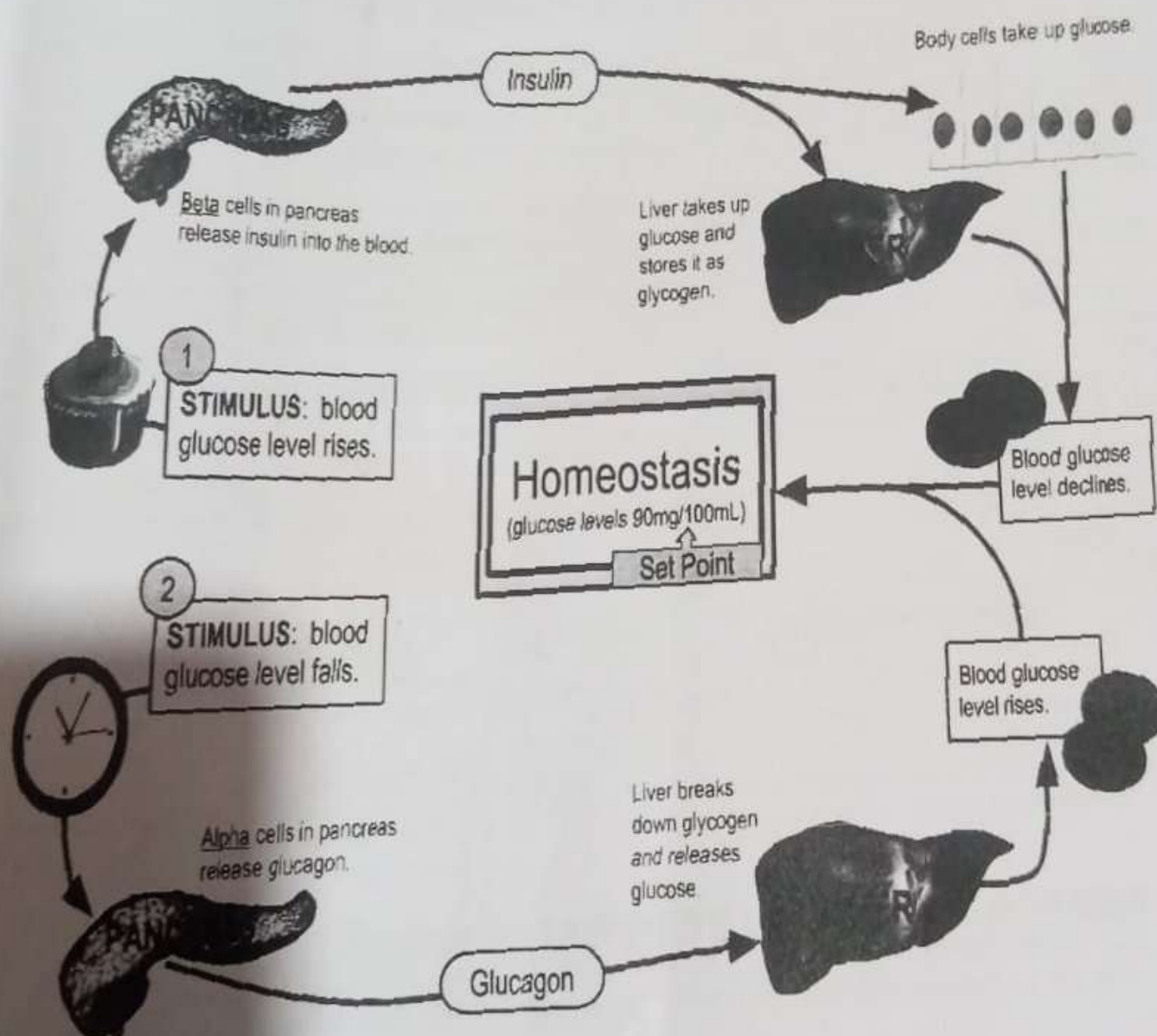
INTER KARACHI

I thank the student who
sent me pictures of the
notes

XII-BIOLOGY NOTES (2023)

HOMEOSTASIS

CHAPTER-15



INTER KARACHI

Chapter
15**HOMEOSTASIS**

DEFINITION: The self maintaining system of the internal stability of the body is known as homeostasis (*Homeo*, like, resembling and *stasis*, standing). This term was introduced by Walter Cannon in 1932. The important aspects of internal environment that must be kept constant are;

1. **OSMOREGULATION:** It is the control of the gain or loss of water and dissolved solutes.
2. **EXCRETION:** It is the disposal of nitrogen containing wastes from a living body.
3. **THERMOREGULATION:** It is the maintenance of internal temperature within a range that allows cells to function efficiently.

FEED BACK SYSTEM: Homeostasis requires a check and balance mechanism operating in the body. This check and balance mechanism is known as feedback system.

TYPES OF FEED BACK SYSTEM: There are two types of feedback system.

Negative feed back: It refers to opposite effect produced in relation to any change in body fluids.

Example:

- (a) Whenever we take any carbohydrate in our diet the level of glucose level is raised in the blood. It triggers the mechanism, which decreases the blood glucose back to normal limit.
- (b) When it is cold out, and body temperature decreases below the set point range. The negative feedback loop will cause the body to shiver producing heat and ultimate body temperature will return within the set point.

Positive feed back: It is concerned with the increase or initiate the change of output for any biological process

Examples:

- (a) During labour when the hormone **oxytocin** stimulates muscular contractions of the uterus, which in turn stimulate the release of more oxytocin.
- (b) Role of platelets for blood clotting at the site of injury.

OSMOREGULATION: It involves maintaining balance between water and solute contents of cell. There are two types of animals on the basis of osmoregulation;

1. Osmoconformers.
2. Osmoregulators.

OSMOCONFORMERS	OSMOREGULATORS
1. Those animals which have an isotonic environment in their body with respect to the surrounding environment are called osmoconformers. 2. They can survive in wide range of salinities. Examples: Marine invertebrates	1. Those animals which have either hypertonic or hypotonic environment inside their body with respect to the surrounding environment are known as osmoregulators. 2. They can survive in a narrow range of salinities. Examples: Marine vertebrates, fresh water and terrestrial animals.

OSMOREGULATION IN FRESH WATER HABITAT:

- Fresh water animals like **fishes** have a hypertonic body-fluids than the surrounding water.
- Thus, they remove excess water by passing large quantities of very dilute urine.

HOMEOSTASIS

BY: PROF RAZA MEHDI

BIOLOGY-XII

- During the excretion process, they lose some essential ions as well. However, this is overcome actively absorbing selected ions from outside, by the help of specialized cells in gills and skin, these cells are known as ionocytes.

OSMOREGULATION IN MARINE HABITAT:

- Marine bony fishes have a hypotonic environment as compare to their external environment.
- Therefore, they have to drink water constantly to compensate the loss of water.
- Inside the body salts are also digested.
- Excessive salts are released by the rectal glands in intestine and salt glands in gills.
- These fishes release small amount of concentrated urine.

OSMOREGULATION IN TERRESTRIAL HABITAT:

Water proof external coverings to avoid dehydration:

- Reptiles, birds and mammals have water proof keratinized epidermis.
- The insects have water proof exoskeleton which is made up of chitin.
- The members of phylum Mollusca have a shell of CaCO_3 .

Release of semi-solid urine: Reptiles, birds and insects excrete uric acid as nitrogenous waste because it requires less amount of water (semi-solid urine) to release from the body.

Use of metabolic water: Some mammals like camel, kangaroo-rat etc. make use of water produced during the breakdown of body fats.

Storing the wastes: Mammals do retain some urea in their kidneys which helps in the re-absorption of water.

EXCRETION IN ANIMALS

In excretion nitrogenous metabolic wastes are released from the body. There are different types of nitrogenous metabolic wastes such as: Ammonia, urea, uric acid, creatinine etc.

AMMONIA (NH_3): Ammonia is a toxic gas. Excreting ammonia by an animal is advantageous due to the reason that its diffusion is rapid across the plasma membrane or body surface and also due to the fact that it does not require energy. However, its removal requires a lot of water for its dilution. Therefore, ammonia is kept as excretory product of those animals which live in hypotonic environment such as fresh water fishes, these animals are called **ammoniotelic**. About 500 ml water is needed to excrete 1 gram of ammonia nitrogen.

UREA ($\text{CO}(\text{NH}_2)_2$): Urea is relatively less soluble in water and about 100,000 times less toxic than ammonia. Urea is produced in the liver by a metabolic cycle that combines NH_3 and CO_2 . In addition to mammals, most amphibians, sharks and some bony fishes also excrete urea, these animals are called **ureotelic**. About 50 ml water is required to release 1 gram of nitrogen in the form of urea.

URIC ACID ($\text{C}_5\text{O}_3\text{N}_4\text{H}_4$): It is a relatively large molecule with extremely low solubility in water and about 10,000 times less toxic than ammonia. Uric acid is excreted by terrestrial animals such as birds, many reptiles, insects. These animals are called **uricotelic**. Only 1ml water is required to release 1 gm nitrogen in the form of uric acid. In human 75% uric acid is excreted by kidneys and 25% is excreted by intestines.

URINARY SYSTEM OF MAN

The urinary system of man consists of the following parts:

- A pair of kidneys
- A pair of ureters
- Urinary bladder and urethra

KIDNEYS:

LOCATION: A pair of kidneys is located against the dorsal body wall in the superior lumbar region. The kidneys extend from T12 – L3. The right kidney is positioned slightly lower than the left.

HOMEOSTASIS

BY: PROF RAZA MEHDI

BIOLOGY

SHAPE AND COLOUR: kidneys are dark red in colour and it is a bean-shaped organ. In the concave side there is a notch called renal sinus.

SIZE: An adult kidney is about 10cm long, 6cm wide and 3cm thick.

RENAL FASCIA: kidneys are enclosed by a thin membranous covering of tough connective tissue called renal fascia.

ADIPOSE CAPSULE: Inside the renal fascia a fatty mass, the adipose capsule surrounds each kidney.

INTERNAL STRUCTURE: The kidney is composed of two distinct regions.

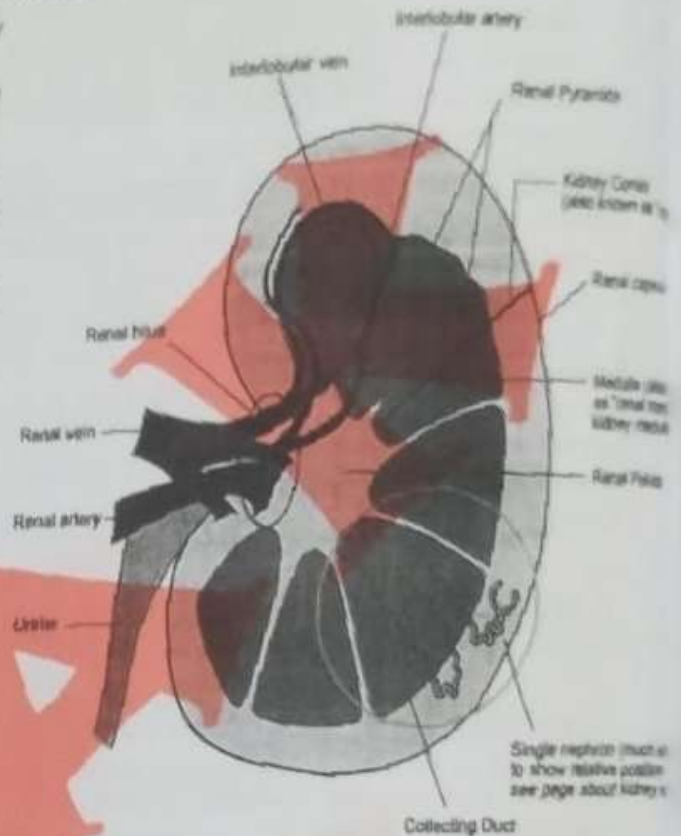
- i. **Renal Cortex:** This is the outer region which is surrounded by a thin layer called renal capsule. The renal cortex contains arterioles, venules and the cells called nephrons. It also produces a hormone **erythropoietin** which helps the production of RBCs.
- ii. **Renal Medulla:** Renal medulla is the inner part of kidney. The medulla has seven to eighteen triangular regions with a striped appearance, the medullary pyramids. The Pyramids are separated by extensions of cortical tissue which are called renal columns. Renal medulla contains the loop of Henle and the collecting ducts of nephrons. On the inner side of medulla is a flat cavity called renal pelvis which continues to the ureter. The extensions of the pelvis are called calyces which form cup shaped areas which enclose the tips of the pyramids and drain it into the pelvis.

FUNCTIONS OF KIDNEYS:

- Kidneys release **nitrogenous wastes**, minerals and extra fluid in the form of urine.
- They also perform **endocrine function** and secrete two hormones i.e., renin and erythropoietin. Renin maintains the blood pressure while the erythropoietin is involved in the production of RBCs.
- They produce **vitamin D3** which helps to absorb calcium and phosphorous in our bones.
- They maintain the pH of our body.
- They regulate water balance by the production of concentrated or dilute urine.

URETERS: From the pelvis a narrow (25 to 30 cm long) tube or ureter arises which brings urine into the urinary bladder.

URINARY BLADDER AND URETHRA: Both the ureters open into a sac-like organ which is called urinary bladder. It stores urine for a short period. Then it passes through a narrow duct which is known as urethra. By the help of urethra, the urine is discharged out of the body.



Abu Huraira Ahmad
Join Our WhatsApp Group

NEPHRONS

Nephrons are the structural and functional unit of kidney. Each kidney consists of about 1 million nephrons.

STRUCTURE OF NEPHRON: A nephron is a tubular structure which is composed of following parts.

1. Malpighian Body:

- It lies in the cortex of kidney.
- It is composed of cup-like Bowman's capsule. Inside this cup lies a dense network of capillaries known as glomerulus.

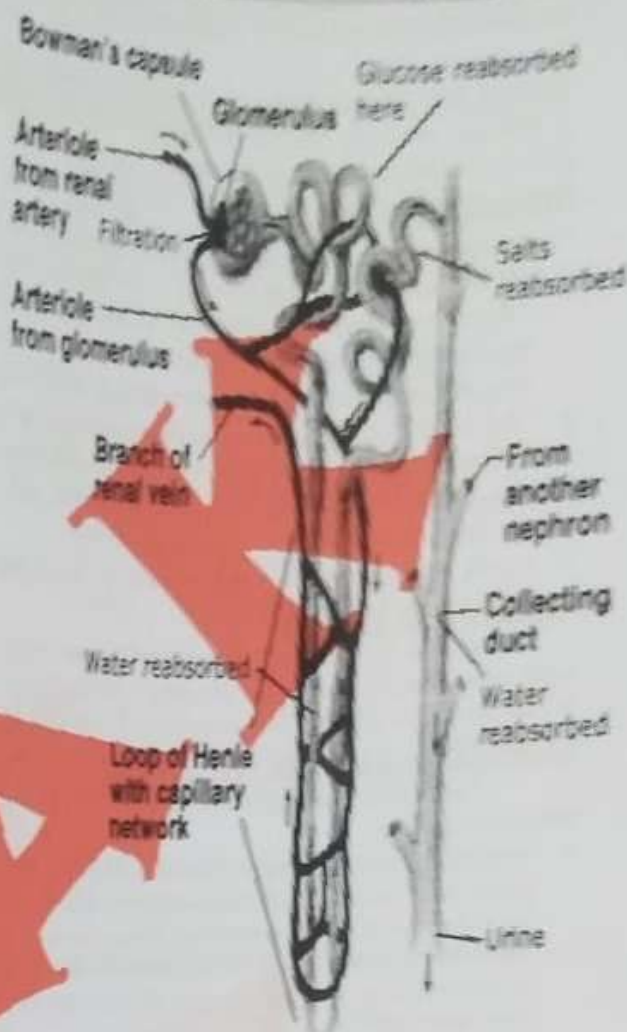
2. Proximal Convoluted Duct (PCT): The Bowman's capsule gives out a coiled tubule known as proximal convoluted tubule. It lies in the cortex region.

3. Loop of Henle: The proximal convoluted duct narrows & descends down into the medulla, makes a U-turn and comes back into the cortex. This narrow U-shaped part is called loop of Henle.

4. Distal Convoluted Duct (DCT): The ascending limb of loop of Henle becomes larger in diameter in the cortex and forms a coil called distal convoluted duct.

5. Collecting duct: The distal convoluted duct finally opens into a common collecting duct which receives urine from a number of nephrons and drains into the pelvis of the kidney.

6. BLOOD VESSELS OF NEPHRON: Blood supply to the glomerulus of a nephron begins as an afferent arteriole arises from renal artery. From the glomerulus blood is carried by the efferent arteriole to the capillary network. One of the network called peritubular capillaries is associated with proximal & distal convoluted ducts while the other are called vasa recta runs straight parallel beside the limbs of loop of Henle.



NEPHRON

REGULATORY FUNCTIONS OF KIDNEY:

Nephrons are the functional units of kidney & they perform regulatory function. This process is completed in the following stages.

GLOMERULAR FILTRATION / ULTRA FILTRATION:

- In the 1st stage non selective ultra filtration of blood occurs while passing through the glomerulus of malpighian body.
- Two factors are responsible for this ultra filtration.

Factor # 1: Both the capillary walls of glomerulus and the walls of Bowman's capsule are sieve like in nature which permits all the molecules except plasma proteins and blood cells to pass through them. The pore size on the blood capillaries is about 10nm in diameter. The inner wall of Bowman's capsule is made up of specialized cells which are known as podocytes. These cells have projections on their surface and in between the projections slit pores or

HOMEOSTASIS

BY: PROF RAZA MEHDI

BIOLOGY

Filtration slits are present. Slit pores are 25 nm wide. The filtered fluid passes through these slits and collected in the nephron.

Factor # 2: Blood pressure in glomerulus is high because the diameter of efferent arteriole is much less than of afferent arteriole. Thus, a hydrostatic pressure is generated which forces the smaller molecules of blood in glomerulus into the Bowman's capsules.

REABSORPTION:

- Active reabsorption of the glomerulus filtrate occurs in the proximal convoluted duct. So, a large amount of filtrate (about 80%) is reabsorbed & returned to circulation. Here glucose (100%), amino acid, vitamins, other organic substances, water (67%), Na^+ , K^+ , and Cl^- (about 50%), Ca^{+2} , Mg^{+2} , and HPO_4^{2-} are reabsorbed.
- Reabsorption of water also occurs when the filtrate from the proximal convoluted duct passes into the descending limb is almost impermeable to other solutes.
- The ascending limb is impermeable to water. Here reabsorption of sodium and chloride ions takes place through active transport. Aldosterone hormone facilitates this process.
- The distal convoluted duct is slightly hypotonic to the blood plasma. In this region concentration of various salts is adjusted under the influence of hormones.
- In the collecting duct of nephron water is reabsorbed in the presence of antidiuretic hormone (ADH).

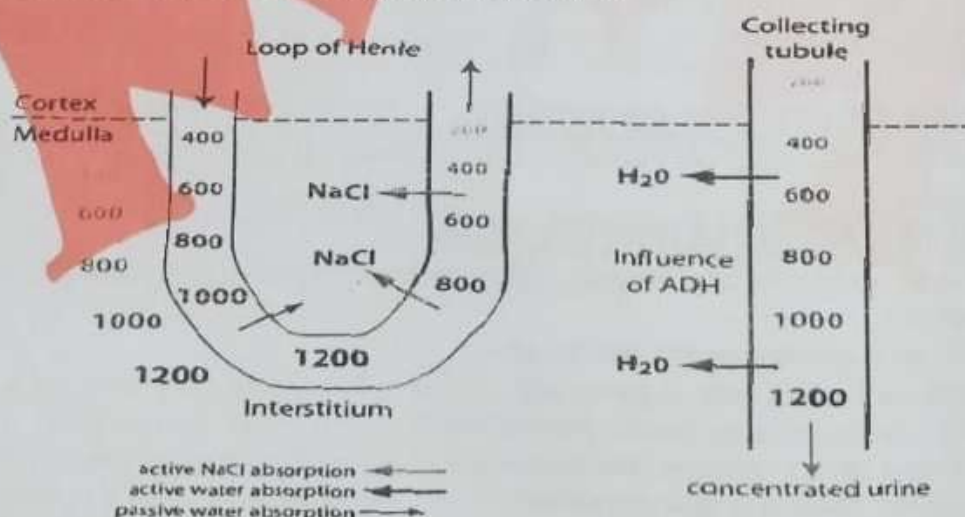
TUBULAR SECRETION:

At the level of both PCT and DCT, some drugs, hormones and nitrogenous substances such as ammonia, urea, uric acid, creatinine are secreted from the blood directly into the filtrate by active transport. This activity is mostly performed by PCT but some of the K^+ ions are also secreted from DCT and collecting duct.

COUNTER - CURRENT EXCHANGE:

There are two counter current systems operating in the medulla which are involved in the exchange of solutes and water.

- One of the counter current is termed as counter current multiplier in the loop of Henle. Ascending limb cells have specialized ionic co-transporter protein, each allow one Na^+ with one K^+ and two Cl^- hence allow lot of ions to move out from the entire limb. These ions are accumulated in medulla and form a hypertonic environment. This environment promotes the passive reabsorption of water from the descending limb. The movement of ions is controlled by aldosterone (steroid) hormone whereas, the reabsorption of water is controlled by antidiuretic hormone (ADH).
- The other counter current system is composed of Vasa recta, the blood vessels that run parallel to the loop of Henle. The blood supply to the vasa recta provides sufficient movement of nutrients and oxygen to the cells of medulla and also takes away the water absorbed from the filtrate.



URINE: After the above-mentioned processes, the filtrate is converted into urine which is transferred to the collecting duct from where it is brought into the renal pelvis.

COMPOSITION OF PLASMA AND URINE

COMPONENTS	% in plasma	% in urine
Water	90.00	95.00
Proteins	8.00	0.00
Glucose	0.1	0.00
Urea	0.03	0.00
Creatinine	0.001	2.0
Ammonia	0.0001	0.075
Uric acid	0.004	0.04
Sodium (Na ⁺)	0.32	0.05
Potassium (K ⁺)	0.01	0.35
Chlorides (Cl ⁻)	0.37	0.15
PO ₄ -3	0.009	0.50
SO ₄ -3	0.002	0.27
		0.18

ANTIDIURETIC HORMONE:

Site of Secretion: Posterior lobe of pituitary gland.

Function of ADH: It increases the reabsorption of water in the collecting duct.

ALDOSTERONE:

Site of Secretion: Adrenal cortex.

Function of aldosterone: It increases the reabsorption of sodium ions in the nephron.

PARATHYROID HORMONE:

Site of Secretion: Parathyroid Glands

Function of parathyroid hormone: It regulates the concentration of calcium ions in blood. It increases the reabsorption of Ca²⁺ in nephrons.

DISORDERS OF URINARY TRACT

1. PYELONEPHRITIS:

Causative agent: *Escherichia coli* (E. coli)

Symptoms: Nausea, vomiting, back pain, fever and shivering.

2. URETHRITIS:

Causative agent: *Escherichia coli* (E. coli), *Chlamydia*, *Neisseria gonorrhea*.

Symptoms: Feeling the frequent or urgent need to urinate, difficulty starting urination, discharge of pus.

3. CYSTITIS:

Causative agent: *Escherichia coli* (E. coli)

Symptoms: A strong urge to urinate, pain and burning sensation during urination, blood in urine, low grade fever.

4. KIDNEY STONES (CALCULUS):

Nephrolithiasis or Urolithiasis: Presence of stone in kidney is called nephrolithiasis.

- Lithonephritis:** The inflammation of kidney due to the irritation of stones is called as Lithonephritis.
- Kidney stones or calculi are mainly (80%) composed of calcium oxalate and 5 to 10% stones are made up of calcium-phosphate or uric acid. Tomato, Spinach, Leafy vegetables are oxalate rich diets which are involved in the production of kidney stones.

Abu Huraira Ahmad

HOMEOSTASIS

BY: PROF RAZA MEHDI

BIOLOGY

- **Infective (struvite) Stones:** About 10% of calculi are termed as infective stone. These calculi consist of combination of Ca, Mg and $\text{NH}_4(\text{PO}_4)$.
- Cystine amino acid may also develop the kidney stones (less than 1%).
- **Causes of kidney stones:**
 - i. **Hypercalciuria:** increase level of calcium in urine.
 - ii. **Hyperoxaluria:** increase level of oxalate in urine.

LITHOTRIPSY (TREATMENT OF kidney stones)

Lithotripsy is a recent method for removing kidney and urethral stones.

PROCEDURE OF LITHOTRIPSY:

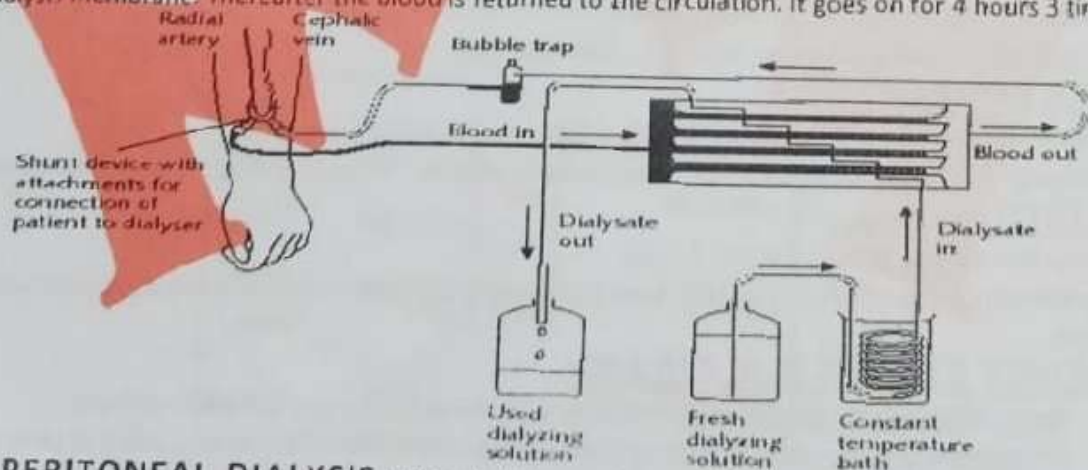
- In **extracorporeal shock wave lithotripsy (ESWL)**, the shock waves are focused on stones from outside the body. After being broken, the small fragments are released from the body by the urine.
- **Percutaneous nephrolithotomy (PCNL)**, in this procedure one catheter is placed in the bladder to drain urine out of the body then another catheter is inserted into the ureter to track down the stone to attached visual device. Once the stone is detected it is crushed and nephroscope grasper is used to pull out the pieces of stone from the kidney.
- In case of very large stones **laparoscopic pyelolithotomy** procedure is used in which the kidneys are cut open to remove stones physically.

RENAL FAILURE: It is a condition in which there is a reduction in the ability of the kidneys to filter waste products from the blood and excrete them in urine.

COMPLICATIONS DUE TO RENAL FAILURE: Due to renal failure nitrogenous wastes start accumulating in the blood which leads to the symptoms such as nausea, loss of appetite, vomiting, weakness, breathlessness, high blood pressure etc. In severe cases it leads to coma and death.

DIALYSIS: It is a technique used to remove waste products from the blood and excess fluids from the body as a treatment for renal failure.

1. **HAEMODIALYSIS:** In this dialysis kidney machine or an artificial kidney containing a fluid (the dialysate) is used. During this dialysis the blood from the **AV fistula** (a plastic tube which is already grafted in between the artery and vein of hand) is collected by a needle and it is pumped repeatedly through a tube of partially permeable membrane (dialysis membrane) bathed in dialysate. While passing through the dialysis membrane the nitrogenous wastes and excess salts are diffused out from the blood into the dialysate through the dialysis membrane. Thereafter the blood is returned to the circulation. It goes on for 4 hours 3 times a week.



2. **PERITONEAL DIALYSIS:** This dialysis requires an abdominal incision. Through the incision a catheter is inserted into the peritoneal cavity. Dialysate from a bag attached to the catheter passes into the cavity where it is left for several hours. During this period waste materials from the vessels are transferred

IAKHI HASSAN

C-1, 14-A, Near Sakhi Hassan, Iqbalwala, Karachi. 850042 - 34889884

NORTH KARACHI

A-955, Sector 11-B, Near U.P. More, North Nazimabad, Karachi. 74346782

HAZIMABAD

I-A, 164, Nazimabad #1, Opposite Govt. College for Men, Karachi. 3832042-1

GULBERG

C-35, Block-13, F.B. Area, Near Water Pump, Chowrang, Karachi. 38330829

MAK SCHOOL SYSTEM

36943080
0341-1248181

MAK WAY GRAMMAR SCHOOL

PRE-PRIMARY CAMPUS 36921059
PRIMARY CAMPUS 34672991
BOYS CAMPUS 0335-4472991, 0311

HEMOSTASIS

into the dialysate. The fluid is then allowed to drain out through the catheter. The process takes about an hour and it is repeated during the day or night.

BIOLOGY-XII

3. **CONTINUOUS RENAL REPLACEMENT THERAPY (CRRT):** This is actually the hemodialysis which is performed in critical kidney failure and it is performed continuously 24 hours a day.

KIDNEY TRANSPLANT: In high degree renal failure also called uremia or end stage disease kidney transplant is the only solution for permanent treatment. Close living relative are sometimes used as donors since this greatly reduces the risk of rejection and a person can live normally with just one kidney. In young patients, having kidney failure transplantation of a healthy kidney is a long-term solution. It is a surgical procedure and successful in 80% of the cases.

Problems associated with kidney transplant:

Human leucocytes antigen system may cause rejection and the transplanted organ is damaged by the immune system of the patient. Therefore, after transplants the immunosuppressant drugs like cyclosporins are given to the patient. Along with these medicines the antibiotics are also prescribed to avoid the infections due to opportunistic parasites. However, after the transplants a patient may suffer from blood clots, bleeding, blockage of ureters, bacterial infections, cancer, heart attack and stroke.

THERMOREGULATION

It is the maintenance of body temperature within a range in which enables the body to function efficiently. The normal temperature range for active life is 10°C to 35°C for most of the organisms.

CLASSIFICATION OF ANIMALS ON THE BASIS OF THERMOREGULATION

There are two methods of heat gain in animal body.

- Externally heat is obtained by solar radiations.
- Internally heat is obtained as a by-product of metabolism and muscular contraction.

POIKILOTHERMS OR COLD BLOODED ANIMALS OR ECTOTHERMS: Poikilotherms are those animals whose body temperature changes according to the environmental temperature. OR Those animals which absorb heat from their environment through behavioral mechanisms are called ectotherms.

Examples: Invertebrates, fishes, amphibian and most reptiles.

HOMEOTHERMS OR WARM BLOODED ANIMALS OR ENDOTHERMS: Homeotherms are those animals which have a constant body temperature which is independent of the environment. OR Those animals that generate their own heat energy to maintain their body temperature are called endotherms.

Examples: Mammals, Birds.

HETEROTHERMS ANIMALS: These animals regulate the body temperature usually as constant, but allows body temperature to fluctuate with the environment when inactive.

Examples: bats and hummingbirds.

MEANS OF THERMOREGULATION IN ANIMALS:

There are two types of thermoregulations.

BEHAVIOURAL REGULATION: Many animals can increase or decrease heat loss by changing their location. This is a type by behavioral regulation.

PHYSIOLOGICAL REGULATION: It is exhibited by slight change in the blood circulation pattern change in the rate of metabolism, cooling by evaporative heat loss and activation of certain muscles.

INTER KARACHI

HOMEOSTASIS

BY: PROF RAZA MEHDI

BIOLOGY-X

THERMOREGULATION IN MAMMALS (MAN)

MECHANISM OF THERMOREGULATION IN COLD TEMPERATURE (Hypothermia):

- Physiological Mechanism
- Behavioural Mechanism

PHYSIOLOGICAL MECHANISM:

A. Non – Shivering Thermogenesis:

1. Erection Of Hair: In furry animals the hair in vertical position (due to the contraction of erector pili muscles) can trap a thick layer of stationary air next to the skin. Due to this, heat loss from the skin is reduced.
2. Reduction in blood flow towards skin: In cold temperature the owing of the arteriole or vaso-constriction occurs so lesser blood is supplied to the body surface to conserve the body heat.
3. Sub-cutaneous fat accumulation: In mammals fat is stored in adipose tissue below the skin. The fat is a bad conductor of heat so it functions as a chemical insulator. This is particularly very important for the survival of aquatic mammals like seals, whales, walrus etc. In these animals a very thick layer of fat called blubber for insulation just under the skin.
4. Coordinating role of brain: The body temperature regulation is based on complex homeostatic system facilitated by feedback mechanism. Hormones also trigger the heat production by increasing the metabolic rate of the body. The metabolic rate is increased for a short term by adrenaline and in long term by the secretion of another hormone, thyroxine. This is known as non-shivering thermogenesis. In some mammals, there is another unique way of thermogenesis due to a specialized tissue called brown fats, found in the neck and between the shoulders. It is specialized for rapid heat production.

B. Shivering Thermogenesis: The reflex action operates when the body temperature begins to drop. The spasmodic (Spasm, violent involuntary muscular contraction) contraction of the muscles is called shivering. This contraction produces heat which helps to raise the body temperature. It is called shivering thermogenesis.

BEHAVIOURAL MECHANISM: Behavioural mechanisms include moving to a warmer place, huddling close together with each other, in humans putting additional clothes.

MECHANISM OF THERMOREGULATION IN HOT TEMPERATURE

(Hyperthermia):

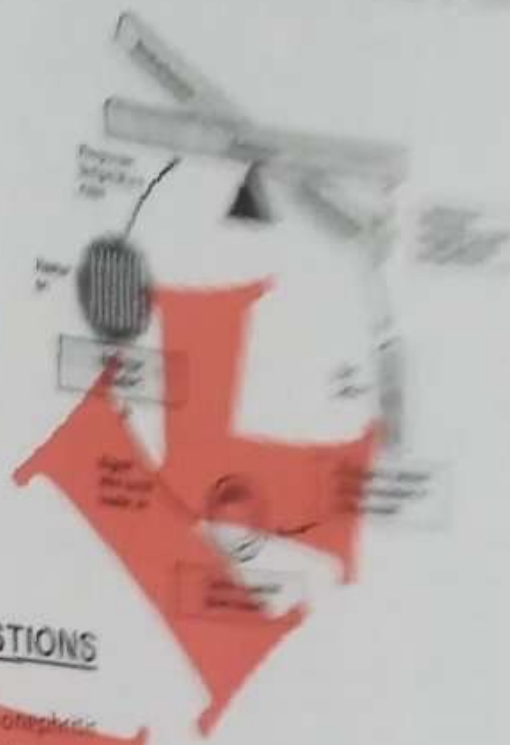
- Physiological mechanism
- Behavioral mechanism

PHYSIOLOGICAL MECHANISM

1. Sweating: Sweating is the production of watery fluid containing dissolved salts from sweat glands in the skin. As sweat evaporates from the skin, it has a cooling effect because the evaporating liquid carries away body heat.
2. Vaso-dilation: It is the expansion of blood vessels i.e. an increase in their diameter so that more blood flows through them. Whenever the body gets too hot vaso-dilation occurs in the dense network capillaries which lie just beneath the epidermis of the skin. Here the blood rapidly loses heat by radiation through the skin, which cools the body.
3. Relaxation of erector pili muscles: By the relaxation of erector pili muscles, the hairs lie more or less flat against the skin. In this position they offer the least possible obstruction to the heat loss by radiation and convection.
4. Panting: It is an efficient method of losing heat from the mouth, tongue and respiratory passage. In dogs panting is the method to release heat for evaporative cooling.
5. Low metabolic rate: The metabolic rate of the body is decreased so less heat is generated by the body.

ROLE OF BRAIN IN TEMPERATURE REGULATION OR THERMOREGULATION

Hypothalamus is the part of brain which has thermoregulatory centre. It is commonly called the thermostat of the body. The hypothalamus is set at a particular temperature called the set point. Any change in the body temperature below or above the set point is detected by the hypothalamus & instructions are sent to appropriate organs to bring the body temperature back to normal.



MULTIPLE CHOICE QUESTIONS

- In high degree renal failure also called
 - Uremia
 - Lithoraphic
 - Nephrolithiasis
 - Cystitis
- The blood clotting is an example of
 - Osmoregulation
 - Negative feed back
 - Positive feed back
 - Feed back system
- The transfer of heat through some physical interaction for example prickly heat power is
 - Conduction heat transfer
 - Convection heat transfer
 - Vasoconstriction
 - Vasodilation
- The transfer of body heat by movement of air or water molecules across the skin is
 - Conduction heat transfer
 - Convection heat transfer
 - Vasoconstriction
 - Vasodilation
- The vasoconstriction may cause
 - Heat loss from the body
 - Renal dysfunction
 - Hypercalciuria
 - Hyperoxaluria
- The reciprocal exchange of Na^+ and K^+ ions from the collecting duct of nephron maintains
 - pH of Urine
 - Water level of urine
 - Volume of urine
 - Blood pressure
- Tubular secretion is mostly performed by
 - PCT
 - DCT
 - Descending limb
 - Ascending limb
- The reabsorption of Na^+ , K^+ and Cl^- ions in ascending limb depends upon
 - Water
 - Co-transporter protein
 - pH of Medulla
 - Vasa recta
- Which of the following statement is incorrect for the ascending limb of loop of henle?
 - It is permeable for Na^+ ions
 - It is permeable for K^+ ions
 - It is permeable for Cl^- ions
 - It is permeable for water
- _____ increases the osmotic gradient in medulla for the reabsorption of water from the nephron.
 - Ammonia
 - Uric acid

HOMEOSTASIS

BY: PROF RAZA MEHDI

BIOLOGY

- c. Urea
11. The marine bony fishes have the salt excretory glands in their _____.
 - a. Gills
 - b. Buccal cavity
 - c. Rectum
 - d. Urinary bladder
12. How much water is reabsorbed from the PCT of nephron?
 - a. 100%
 - b. 67%
 - c. 80%
 - d. 87%
13. Mammals store _____ in their kidneys for the reabsorption of water.
 - a. Ammonia
 - b. Creatinin
 - c. Urea
 - d. Uric acid
14. Name the excretory material in insects, birds and reptiles.
 - a. Ammonia
 - b. Creatinin
 - c. Urea
 - d. Uric acid
15. Renin is a hormone produced by the kidneys which controls _____.
 - a. Calcium levels in blood
 - b. RBCs production
 - c. Blood pressure
 - d. Potassium level in blood
16. Which is not a homeostatic organ in our body?
 - a. Skin
 - b. Heart
 - c. Liver
 - d. Kidneys
17. Erythropoietin is a hormone produced by the kidneys which controls _____.
 - a. Calcium levels in blood
 - b. RBCs production
 - c. Blood pressure
 - d. Potassium level in blood
18. Which is not the function of kidneys?
 - a. Homeostatic organ
 - b. Endocrine organ
 - c. Vitamin D3 production
 - d. Maintains the pH of our body
 - e. None of these
19. The inner wall of the bowman's capsule is lined with _____.
 - a. Pinacocytes
 - b. Porocytes
 - c. Choanocytes
 - d. Podocytes
20. The thermoregulatory centre is found in _____ of brain.
 - a. Hypothalamus
 - b. Cerebral hemisphere
 - c. Limbic system
 - d. Cerebrum

ANSWERS KEY

1.	2.	3.	4.	5.
a	b	a	b	b
6.	7.	8.	9.	10.
a	a	b	d	c
11.	12.	13.	14.	15.
a	b	c	d	c
16.	17.	18.	19.	20.
b	b	e	d	a

Abu Huraira Ahmad

SAKHI HASSAN

C-1, 14-A, Near Sakhi Hassan Chowrang, Karachi.
36943080 - 36943080

NORTH KARACHI

A-555, Sector 11-B, Near U.P. North, North Karachi, Karachi.
36947788

NAZIMABAD

1-A, 104, Nazimabad # 1, Opposite Govt. College for Men, Karachi.
36929042-3

GULBERG

C-35, Block-12, F. B. Area, Near Water Pump Chowrang, Karachi.
36360939

MAK SCHOOL SYSTEM

36943080
0341-1248181

MAK WAY GRAMMAR

PRE-PRIMARY CAMPUS
36921059

PRIMARY CAMPUS
36671891

BOYS CAMPUS
0341-1248181

----- OUR CHANNELS -----

INTER KARACHI (BIEK)

[Channel Link](#)

MDCAT --- Material & Guidance Group

[Channel Link](#)

ECAT --- Preparation Group

[Channel Link](#)

----- OUR GROUPS -----

INTER KARACHI

[Group Link](#)

- This group provides basic news, information, and study materials. Phone numbers will not be visible to other members.

2nd YEAR

[Group Link](#)

- This group serves the same purpose as the previous one. You can vote in this group to express your preferences.

2nd YEAR (2)

[Group Link](#)

- This group is identical to the previous one. Please join only one group from this set.

Inter Discussion (Boys)

[Group Link](#)

- In this group, you can engage in discussions and share your thoughts and questions. Please note that this group is exclusively for males.

MDCAT --- Material & Guidance Group

[Group Link](#)

- Created for MDCAT test preparation and guidance for the 2024 session. Start your preparation from the intermediate level. Group details are available in the description.

ECAT --- Preparation Group

[Group Link](#)

- Established for ECAT test preparation and guidance for the 2024 session. Begin your preparation from the intermediate level. Further information is provided in the group description.

Admin:

Abu Huraira Ahmad